```

<!doctype html>
<html lang="en">
  <head>
    <!-- Required meta tags -->
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">

    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOmLASjC" crossorigin="anonymous">

    <title>Game Simulator</title>
    <style>
      body {
        background-color: #f5f5f5;
        color: #333;
        font-family: "Segoe UI", Arial, sans-serif;
      }
      h1, h2, h3 {
        color: #2e7d32; /* green accent */
      }
      .navbar {
        background-color: #263238 !important; /* deep navy */
      }
      .navbar-brand, .navbar-nav .nav-link {
        color: #ffffff !important;
      }
      .section {
        margin-top: 2rem;
        margin-bottom: 2rem;
        padding: 1rem 1.5rem;
        background: #ffffff;
        border-radius: 10px;
        box-shadow: 0px 2px 6px rgba(0,0,0,0.1);
      }
```

```
      .bio-img {
        max-width: 250px;
        border-radius: 50%;
        margin-bottom: 1.5rem;
        border: 4px solid #2e7d32;
      }
      hr {
        border-top: 2px solid #2e7d32;
        opacity: 0.6;
      }
      a {
        color: #0288d1;
        text-decoration: none;
      }
      a:hover {
        text-decoration: underline;
      }
    </style>
  </head>
  <body>
    <nav class="navbar navbar-expand-md">
      <a class="navbar-brand" href="#">BGF Analytics
  </a>
      <button class="navbar-toggler" type="button"
  data-bs-toggle="collapse" data-bs-target="#navbarNav"
  >
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="
  navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link"
  href="{{ url_for('home_page')}}">Home Page</a></li>
          <li class="nav-item"><a class="nav-link"
  href="{{ url_for('resume')}}">Resume</a></li>
          <li class="nav-item"><a class="nav-link"
  href="{{ url_for('projects')}}">Projects</a></li>
          <li class="nav-item"><a class="nav-link"
  href="{{ url_for('game_sim')}}">Game Simulator</a></
  li>
          <li class="nav-item"><a class="nav-link"
```

```
67 href="{{ url_for('XGB_sim')}}">Bracket Simulator</a
   ></li>
68         </ul>
69       </div>
70     </nav>
71     <h1>March Madness Game Simulator</h1>
72         <p>
73             Welcome to the March Madness Bracket
   Simulator!
74         </p>
75         <p>
76             Set the Simualation Controls to your
   liking and the system will simulate all games from
   from the First Four to the Championship.  Good Luck!
77         </p>
78     <div class="container mt-4">
79     <form id="simulationForm" method="POST" onsubmit
   ="return false;">
80
81      <!-- User Feature Controls Description -->
82         <div class="card bg-dark text-white mb-4">
83           <div class="card-body">
84             <h5 class="card-title">Simulation
   Controls Explained</h5>
85
86             <div class="row">
87               <div class="col-md-6 mb-3">
88                 <p><strong>Chaos Level (Noise Std):
   </strong> Controls randomness in score predictions.
89                   <br>• <em>Low (0.5)</em>: Very
   stable outcomes, few upsets.
90                   <br>• <em>Medium (2.5)</em>:
   Balanced realism and unpredictability.
91                   <br>• <em>High (5.0)</em>: Wild
   swings, frequent upsets and score volatility.
92                 </p>
93               </div>
94
95               <div class="col-md-6 mb-3">
96                 <p><strong>Seed Consistency Bias:</
   strong> Adjusts how much top seeds resist randomness
```

```
.
                  <br>• <em>Low (0.5)</em>: Top seeds dominate, fewer surprises.
                  <br>• <em>Medium (0.8)</em>: Realistic balance of consistency and chaos.
                  <br>• <em>High (1.2)</em>: All seeds treated more equally, more parity.
                </p>
              </div>

              <div class="col-md-6 mb-3">
                <p><strong>Model vs History:</strong> Blends predictive model with historical seed win rates.
                  <br>• <em>Low (0.0)</em>: Pure model logic, no historical bias.
                  <br>• <em>Medium (0.5)</em>: Balanced between model and history.
                  <br>• <em>High (1.0)</em>: Pure historical outcomes, less model influence.
                </p>
              </div>

              <div class="col-md-6 mb-3">
                <p><strong>Normalize Scores by Round:</strong> Aligns scores with historical round averages.
                  <br>• <em>Yes</em>: Scores reflect typical round scoring (e.g. lower in Final Four).
                  <br>• <em>No</em>: Raw model outputs used, may be inflated or deflated.
                </p>
              </div>
            </div>
          </div>
        </div>

      <!-- User Simulation Varible Contols -->
      <div class="row mb-3">
         <div class="col-md-3">
            <label for="noise_std" class="form-label
```

```
">Chaos Level (Noise Std)</label>
                <select class="form-select" id="noise_std" name="noise_std">
                 {% for val in [0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0] %}
                    <option value="{{ val }}">{{ val }}</option>
                    {% endfor %}
                </select>
      </div>

      <div class="col-md-3">
        <label for="consistency_multiplier" class="form-label">Seed Consistency Bias</label>
           <select class="form-select" id="consistency_multiplier" name="consistency_multiplier">
             {% for val in [0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2] %}
               <option value="{{ val }}">{{ val }}</option>
             {% endfor %}
           </select>
      </div>

      <div class="col-md-3">
        <label for="blend_weight" class="form-label">Model vs History</label>
        <select class="form-select" id="blend_weight" name="blend_weight">
          {% for val in [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0] %}
            <option value="{{ val }}">{{ val }}</option>
          {% endfor %}
        </select>
      </div>

      <div class="col-md-3">
        <label for="apply_normalization" class="form-label">Normalize Scores by Round</label>
```

```
        <select class="form-select" id="apply_normalization" name="apply_normalization">
          <option value="True">Yes</option>
          <option value="False" selected>No</option>
        </select>
      </div>
    </div>
     <!-- Run Simulation Button -->
    <button id="simulateBtn" type="submit" class="btn btn-primary mt-3">Simulate Bracket</button>
  </form>

  <!-- Champion Display Area (ADDED THIS!) -->
  <div id="champion-output" class="alert alert-success mt-4" style="display: none; font-size: 1.2em; font-weight: bold;"></div>

<!-- Fetch User Inputs -->
<script>
    document.addEventListener("DOMContentLoaded", () => {
        const btn = document.getElementById('simulateBtn');
        const championDiv = document.getElementById("champion-output");
        const resultArea = document.getElementById("resultArea");

        btn.addEventListener('click', function (e) {
            e.preventDefault();

            // Show loading state
            btn.disabled = true;
            btn.textContent = "Simulating...";
            championDiv.style.display = "none";
            resultArea.innerHTML = '<div class="alert alert-info">Running simulation...</div>';

            const formData = new FormData(document.getElementById('simulationForm'));

```

```
            fetch("/simulate_bracket", {
                method: "POST",
                body: formData
            })
            .then(res => {
                console.log("Status:", res.status);
                if (!res.ok) {
                    throw new Error(`HTTP ${res.status}: ${res.statusText}`);
                }
                return res.json();
            })
            .then(data => {
                console.log("Received data:", data);

                if (data.error) {
                    throw new Error(data.error);
                }

                if (data.champions && data.champions.length > 0) {
                    const [name, seed] = data.champions[0];
                    championDiv.textContent = `🏆 Champion: ${name} (#${seed})`;
                    championDiv.style.display = "block";
                }

                if (data.logs && data.logs.length > 0) {
                    resultArea.innerHTML = `<pre class="bg-dark text-white p-3 rounded">${data.logs[0]}</pre>`;
                } else {
                    resultArea.innerHTML = '<div class="alert alert-warning">No simulation logs received.</div>';
                }
            })
            .catch(err => {
```

```
                console.error("Fetch failed:", err);
                championDiv.style.display = "none";
                resultArea.innerHTML = `<div class="alert alert-danger">Simulation failed: ${err.message}</div>`;
            })
            .finally(() => {
                // Reset button state
                btn.disabled = false;
                btn.textContent = "Simulate Bracket";
            });
        });
    });
</script>

<!-- Results Area -->
    <div id="resultArea" style="margin-top: 1em; font-weight: bold;"></div>
</div>
    <!-- Optional JavaScript; choose one of the two! -->

    <!-- Option 1: Bootstrap Bundle with Popper -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bundle.min.js" integrity="sha384-MrcW6ZMFYlzcLA8Nl+NtUVF0sA7MsXsP1UyJoMp4YLEuNSfAP+JcXn/tWtIaxVXM" crossorigin="anonymous"></script>

    <!-- Option 2: Separate Popper and Bootstrap JS -->
    <!--
    <script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.9.2/dist/umd/popper.min.js" integrity="sha384-IQsoLXl5PILFhosVNubq5LC7Qb9DXgDA9i+tQ8Zj3iwWAwPtgFTxbJ8NT4GN1R8p" crossorigin="anonymous"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.min.js" integrity="sha384-cVKIPhGWiC2Al4u+LWgxfKTRIcfu0JTxR+EQDz/
```

```
bgldoEyl4H0zUF0QKbrJ0EcQF" crossorigin="anonymous
"></script>
    -->
  </body>

</html>
```